ELK

ELASTIC SEARCH,

LOGSTASH,

KIBANA

What is the Elastic Stack?

What is the Elastic Stack?

Elasticsearch

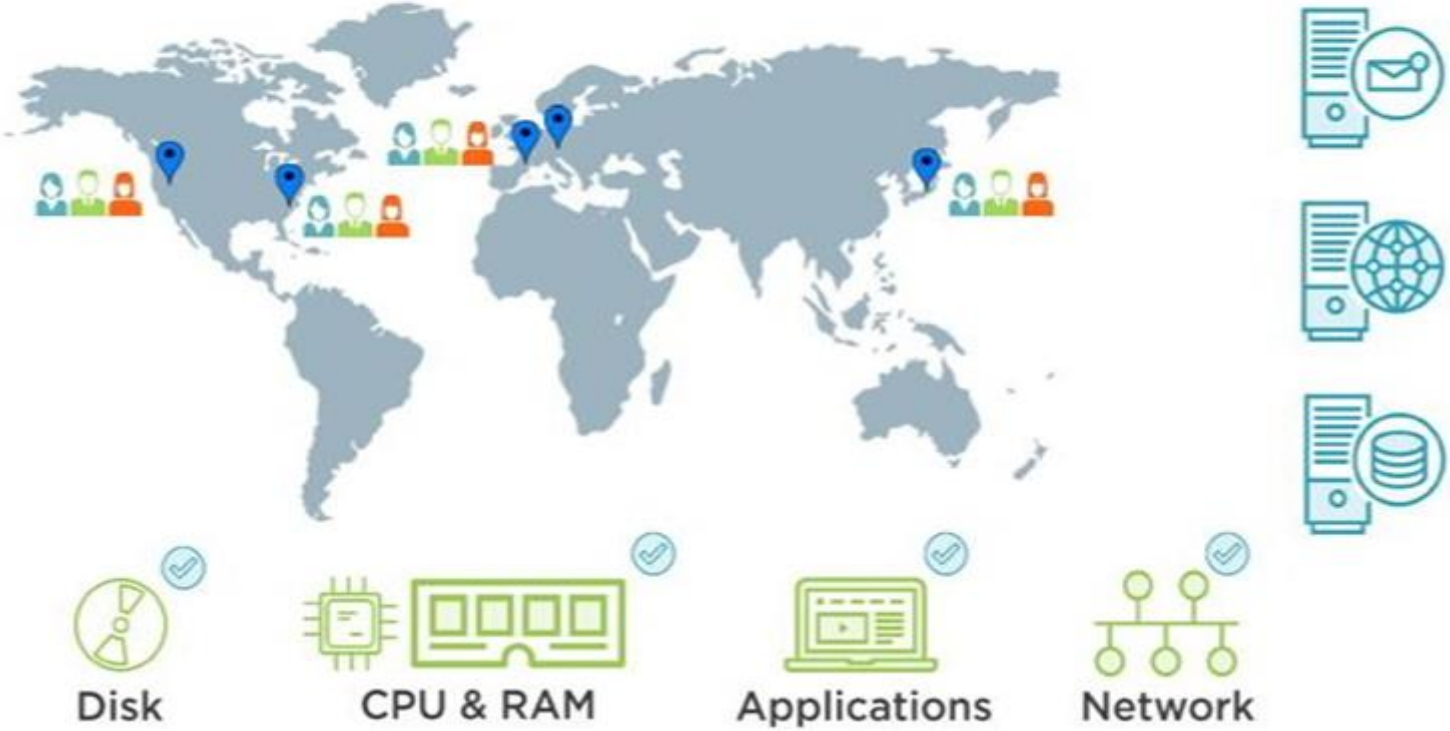
Logstash

Beats

Kibana

- ✓ Free
- ✓ Open Source
- ✓ Great at full-text searching

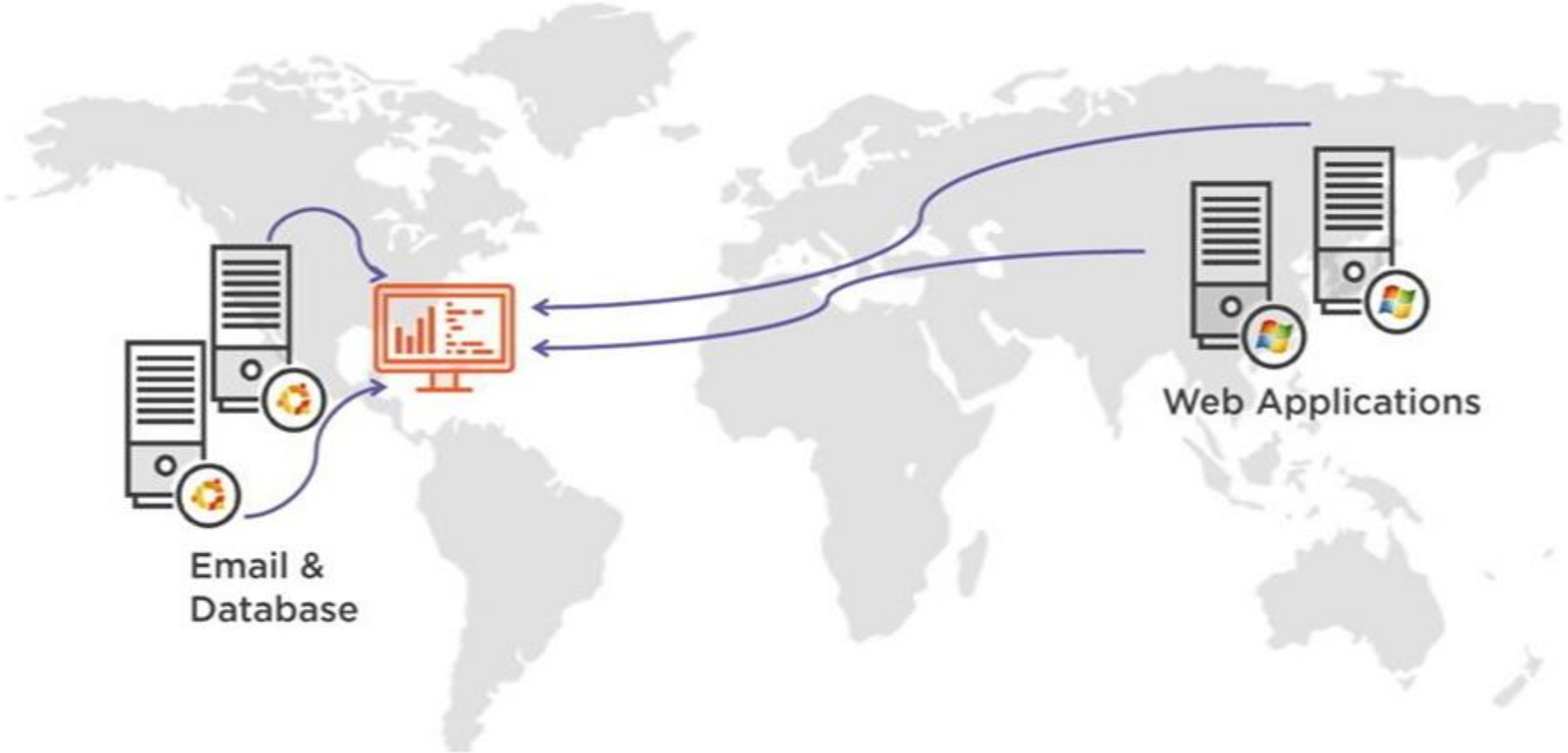
Globomantics Is Worldwide



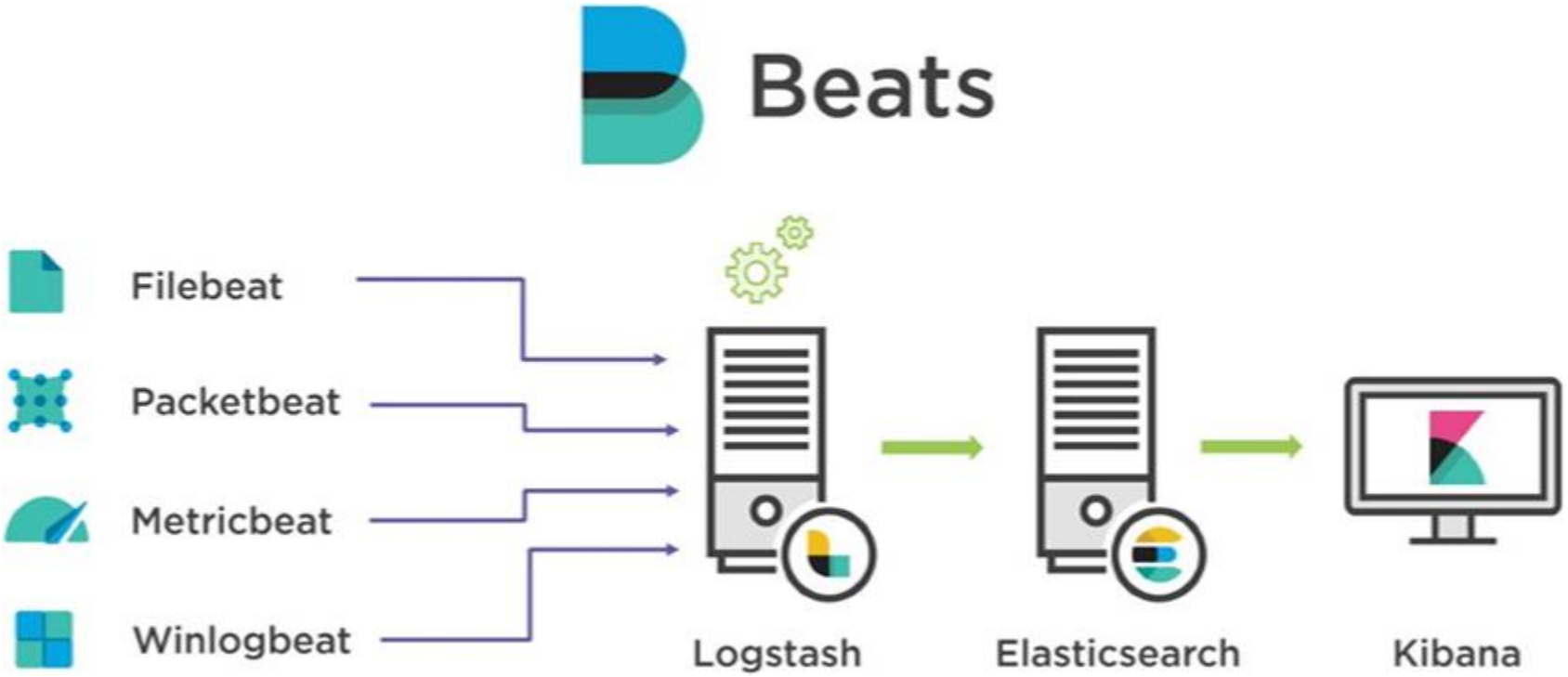
Your Role Today

DevOps / IT

GLOBOMANTICS

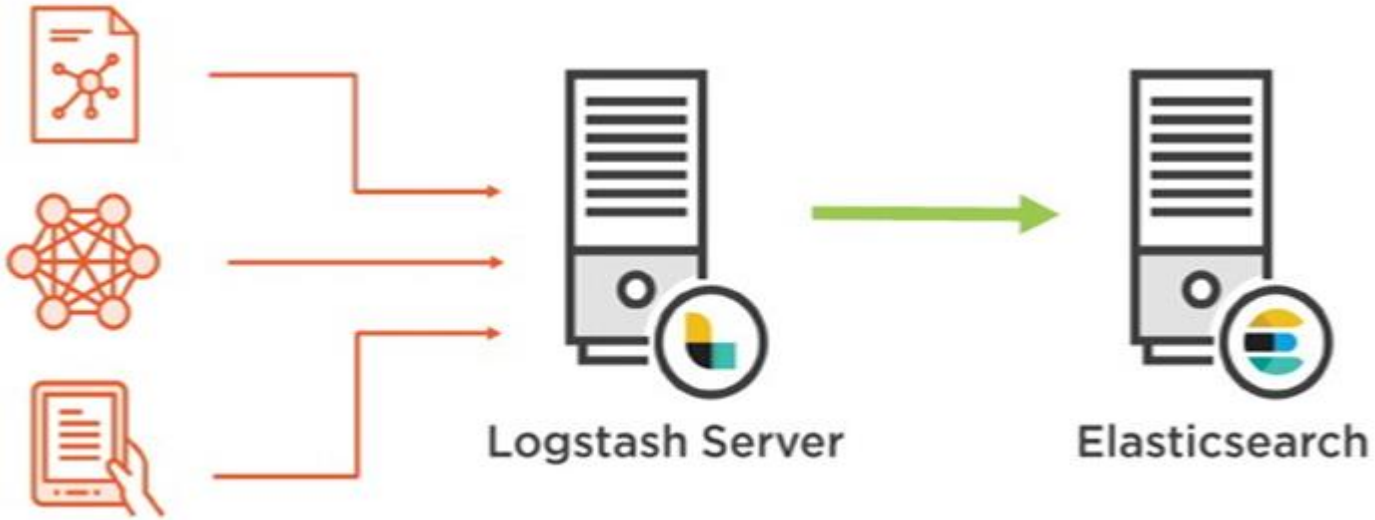

Email &
Database
Web Applications

US EAST


Beats
Filebeat
Packetbeat
Metricbeat
Winlogbeat
Logstash
Elasticsearch
Kibana

Logstash Is a Data Collection Engine

1. Ingest
2. Enhance or modify
3. Forward



Logstash Configuration

```
input {
}
```

Where is data coming from? Logs? Beats?

```
filter {
}
```

How should we parse the data? Ignore some? Modify any?

```
output {
}
```

Where should we store the logs? Back end? Elasticsearch?

Beats

Small, lightweight utilities for reading logs from a variety of sources. Usually sends data to Logstash

Filebeat: Text log files

Metricbeat: OS and applications

Packetbeat: Network monitoring

Winlogbeat: Windows Event log

Libbeat: Write your own

Logstash

Aggregates, filters, and supplements log data

Forwards altered logs to Elasticsearch

Sending logs directly to Elasticsearch without Logstash can lead to inconsistent data

Elasticsearch

Distributed, fast, highly scalable document database

Created by Shay Banon in 2010

We'll use a simple single node cluster

Kibana

Web-based front-end

Works easily with Elasticsearch for charts, graphs, and visualizing data

Free from the Elastic company

kibana

Management / Kibana

Index Patterns Saved Objects Advanced Settings

- Discover
- Visualize
- Dashboard
- Timelion
- Dev Tools
- Management

Warning
No default index pattern. You must select or create one to continue.

Configure an index pattern

In order to use Kibana you must configure at least one index pattern. Index patterns are used to identify the Elasticsearch index to run search and analytics against. They are also used to configure fields.

Index name or pattern

logstash-*

Patterns allow you to define dynamic index names using * as a wildcard. Example: logstash-*

Time Filter field name refresh fields

☐ Expand index pattern when searching [DEPRECATED]

With this option selected, searches against any time-based index pattern that contains a wildcard will automatically be expanded to query only the indices that contain data within the currently selected time range.

Searching against the index pattern *logstash-** will actually query Elasticsearch for the specific matching indices (e.g. *logstash-2015.12.21*) that fall within the current time range.

With recent changes to Elasticsearch, this option should no longer be necessary and will likely be removed in future versions of Kibana.

☐ Use event times to create index names [DEPRECATED]

Time Filter field name is required

Installation of Elsaticsearch

Installing ElasticSearch

1. Launch the ubuntu 16 Server with 4gb ram
2. install jdk like below:

   apt-get install openjdk-8-jre-headless

1. Check the version

   java -version

1. Create a directory

   mkdir myelk

   cd myelk

Installing ElasticSearch

5. download elastic search package
   wget
   https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-5.6.3.deb
6. dpkg -i elasticsearch-5.6.3.deb

Installing ElasticSearch

7. Open vi /etc/elasticsearch/elasticsearch.yml and do the changes as below:

   #cluster.name : globo-clustering(give any name)
   #Node.name : public DNS of Elasticsreach(in Production private DNS)
   #Network.host : private ip of elasticsreach
   #Http.port : 9200

Note: uncomment the # symbol

8. Save the file

Installing ElasticSearch

9. Increase the memory map count by
   sysctl -w vm.max_map_count=262144
9. Restart elasticsearch cluster service by
   service elasticsearch restart

9. Open browser and execute below:

   http://<ipadress>:9200

Installation of Logstash

Installing Logstash

1. Launch instance of Ubuntu 16.04 4gb ram with port no.5044
2. apt-get update
3. Install java

   apt-get install openjdk-8-jre-headless

1. wget -qO - https://packages.elastic.co/GPG-KEY-elasticsearch | sudo apt-key add -

Installing Logstash

5. echo "deb https://artifacts.elastic.co/packages/5.x/apt stable main" | sudo tee -a /etc/apt/sources.list.d/elasticsearch-5.x.list
6. apt-get update
7. apt-get install logstash
8. cd /usr/share/logstash

Installing Logstash

9. Now execute below command

   bin/logstash -e "input { stdin {} } output { stdout {} }"

9. Press enter

10. enter some value.

11. service logstash restart

Beats configuration in logstash

Beats configuration in Logstash

1. Cd /etc/logstash/conf.d/
2. Vi beats.conf

```
input {
  beats {
    port => 5044
  }
}
```

Beats configuration in Logstash

```
output {
  elasticsearch {
    hosts => "<elasticsearchpublicip>:9200"
    manage_template => false
    index => "%{[@metadata][beat]}-
%{[@metadata][version]}-%{+YYYY.MM.dd}"
    document_type => "%{[@metadata][type]}"
  }
}
```

Beats configuration in Logstash

3. Save the file
4. Service logstash restart
5. Navigate to cd /usr/share/logstash
6. Type the command below:
   bin/logstash -f /etc/logstash/conf.d/

Installation of Kibana

Installing Kibana

1. Launch instance of Ubuntu 16.04 2gb ram with port no.5601
2. apt-get update
3. wget -qO - https://packages.elastic.co/GPG-KEY-elasticsearch | sudo apt-key add -

Installing Kibana

4. echo "deb https://artifacts.elastic.co/packages/5.x/apt stable main" | sudo tee -a /etc/apt/sources.list.d/elasticsearch-5.x.list
5. apt-get update
6. apt-get install kibana

Installing Kibana

7. vi /etc/kibana/kibana.yml

   server.host: private <ipaddress> of kibina

   server.name: private <hostname> of kibina

   elasticsearch.url: <elasticsearchurl> of electric search

8. Save the file

9. service kibana restart

10. Open browser and execute below:

    http://<kinbapublicip>:5601

Installation of Filebeats

Installing Filebeats

1. Launch amazon linux2 machine
2. apt-get update
3. curl -L -O https://artifacts.elastic.co/downloads/beats/filebeat/filebeat-5.6.4-x86_64.rpm
4. sudo rpm -vi filebeat-5.6.4-x86_64.rpm
5. cd /etc/filebeat/
6. Vi filebeat.yml

Installing Filebeats

#tags: ["ap-southeast-1"]

#fields:

globo_environment: production(add this line below fileds)

Note:uncomment # symbol

output.elasticsearch

hosts

Note: comment with # symbol above two

Installing Filebeats

#output.logstash

#hosts: ["<public ip of logstash>:5044"]

Note: uncomment # symbol

8. Save the file

Installing Filebeats

9. curl -H 'Content-Type: application/json' -XPUT 'http://<elasticsearch public ip>:9200/_template/filebeat' -d@/etc/filebeat/filebeat.template.json
10. sudo /etc/init.d/filebeat restart

Kibana Dashboard

1. Copy and paste the public ip of public ip kibana instance
2. Open browser execute like below: <public ip of kibana>:5601
1. In search bar type filebeat-*
2. Click on discover select filebeat-*
3. Select timestamp ie., @timestamp
4. Check the index pattern

kibana

214,397 hits

New Save Open Share Last 15 minutes

*

Discover
Visualize
Dashboard
Timelion
Management
Dev Tools

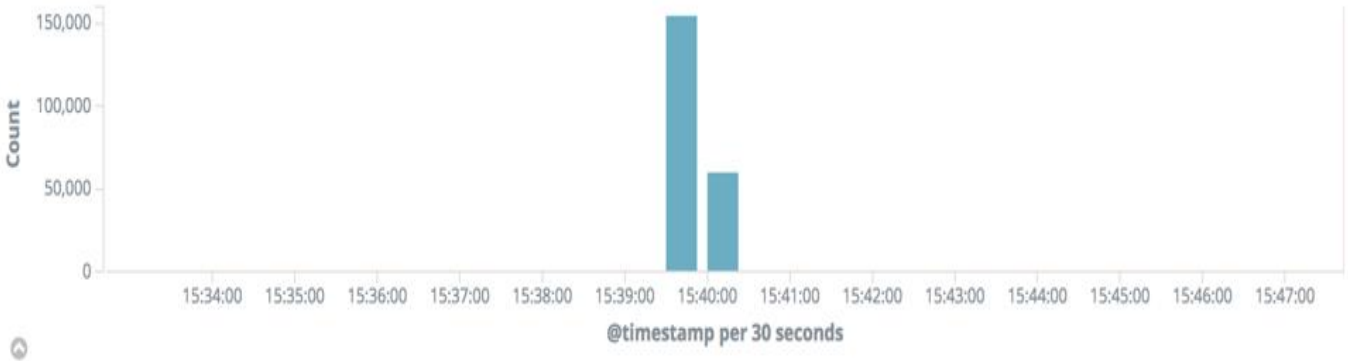
filebeat-*

Selected Fields

Available Fields

@timestamp
_id
_index
_score
_type
beat.hostname
beat.name
input_type
message
offset
source
type

October 4th 2016, 15:32:42.769 - October 4th 2016, 15:47:42.769 — by 30 seconds



Time	_source
October 4th 2016, 15:45:38.663	{ "@timestamp": "2016-10-04T13:45:38.663Z", "beat": { "hostname": "Rhodas-MacBook-Pro.local", "name": "Rhodas-MacBook-Pro.local" }, "input_type": "log", "message": "Oct 4 15:45:32 Rhodas-MacBook-Pro.local mds[55]: (DiskStore.Normal:2382) 8001 1.000006", "offset": 342947, "source": "/private/var/log/system.log", "type": "log" }
October 4th 2016, 15:40:38.619	{ "@timestamp": "2016-10-04T13:40:38.619Z", "beat": { "hostname": "Rhodas-MacBook-Pro.local", "name": "Rhodas-MacBook-Pro.local" }, "input_type": "log", "message": "Oct 4 15:40:29 (68.44) AuthenticationAllowed completed: record \"dedemorton\", result: Success (0).", "offset": 1584, "source": "/private/var/log/accountpolicy.log", "type": "log" }

Installation of winlogbeats on windows

Install winlogbeats

1. Google it install winlogbeat and click on first link
2. Download the Winlogbeat zip file from the downloads page.
3. Click on download page link
4. Click on windows 64 bit or windows 32 bit
5. Zip file will be downloaded
6. Extract the contents into C:\Program Files
7. Rename the winlogbeat-<version> directory to Winlogbeat

Installing winlogbeats

8. Open vi winlogbeat.yml with nodepad ++ and do the changes as below:

   #tags: ["ap-southeast-1"]

   #fields:

   globo_environment: production(add this line below Fields section)

Note:uncomment # symbol

output.elasticsearch

hosts

Note: comment with # symbol above two

Installing winlogbeats

#output.logstash

#hosts: ["<public ip of logstash>:5044"]

Note: uncomment # symbol

9. Save the file

Installing winlogbeats

10. Open a PowerShell prompt as an Administrator (right-click on the PowerShell icon and select Run As Administrator).
11. From the PowerShell prompt, run the following commands to install the service.

```
PS C:\Users\Administrator> cd 'C:\Program Files\Winlogbeat'
```

```
PS C:\Program Files\Winlogbeat> .\install-service-winlogbeat.ps1
```

Installing winlogbeats

12. To export the index template, run:

```
PS > .\winlogbeat.exe export template --es.version 7.3.0 | Out-File -Encoding UTF8 winlogbeat.template.json
```

12. To install the template, run:

```
PS > Invoke-RestMethod -Method Put -ContentType "application/json" -InFile winlogbeat.template.json -Uri http://<elasticsearch public ip>:9200/_template/winlogbeat-7.3.0
```

Installing winlogbeats

14. From the PowerShell prompt, change to the directory where we installed Winlogbeat, and run:

```
PS > .\winlogbeat.exe setup --dashboards
```

14. From the PowerShell prompt, change to the directory where we installed Winlogbeat, and run:

Installing winlogbeats

```
PS > .\winlogbeat.exe setup -e `
  -E output.logstash.enabled=false `
  -E output.elasticsearch.hosts=['<elasticsearch
    Public ip>:9200'] `
  -E output.elasticsearch.username=winlogbeat_internal `
  -E output.elasticsearch.password=OUR_PASSWORD `
  -E setup.kibana.host=<kibana public ip>:5601
```

Installing winlogbeats

16. Start service using

```
PS C:\Program Files\Winlogbeat> Start-Service winlogbeat
```

Kibana Dashboard

1. Copy and paste the public ip of public ip kibana instance
2. Open browser execute like below: <public ip of kibana>:5601
1. In search bar type winlogbeat-*
2. Click on discover select winlogbeat-*
3. Select timestamp ie., @timestamp
4. Check the index pattern

54.255.170.251:920 | EC2 Management | My Drive - Google | ELK Instalation - G | ELKZONE/beats.co | Parsing Logs with | Kibana

54.169.238.188:5601/app/kibana#/discover?_g=()&_a=(columns:!(_source),index:AV_aol1w3rb6XUZsnssy,interval:auto,query:(match_all:()),sort:!('@timestamp',desc))

Apps | NR Digital | EC2 Management Co | WhatsApp Web

kibana

Discover
Visualize
Dashboard
Timelion
Dev Tools
Management
Collapse

12 hits

New | Save | Open | Share | Last 15 minutes

Search... (e.g. status:200 AND extension:PHP) — Uses lucene query syntax

Add a filter +

winlogbeat-*

Selected Fields

? _source

Available Fields

@timestamp
t @version
t _id
t _index
_score
t _type
t beat.hostname
t beat.name
t beat.version
t computer_name
t event_data.Binary

November 20th 2017, 23:31:17.466 - November 20th 2017, 23:46:17.466 — Auto

Time	_source
November 20th 2017, 23:40:11.000	computer_name: MAHA-PC keywords: Classic log_name: Application level: Information record_number: 6883 event_data.param1: 3170 event_data.param2: 0 message: Content successfully updated. Major Version: 3170 Minor Version: 0 type: wineventlog tags: ap-southeast-1, beats_input_codec_plain_applied @timestamp: November 20th 2017, 23:40:11.000 event_id: 5,008 @version: 1 beat.name: MAHA-PC beat.hostname: MAHA-PC beat.version: 5.6.3
November 20th 2017, 23:40:11.000	computer_name: MAHA-PC keywords: Classic log_name: Application level: Information record_number: 6884 event_data.param1: 3170 event_data.param2: 0 message: McShield successfully started. Major Version: 3170 Minor Version: 0 type: wineventlog tags: ap-southeast-1, beats_input_codec_plain_applied @timestamp: November 20th 2017, 23:40:11.000 event_id: 5,000 @version: 1 beat.name: MAHA-PC beat.hostname: MAHA-PC beat.version: 5.6.3

11:46 PM
11/20/2017